

ECT 250: Survey of e-commerce technology

Introduction to e-commerce



Introduction to the course

- Instructor for ECT 250-603: [Amber Settle](#)
- ECT 250 gives a survey of the key technological elements of e-commerce and provides insight into e-commerce infrastructures. It also covers some business strategies essential to e-commerce. The web authoring tool FrontPage will be used.

A quick survey

- Which of the following have you done?
 - Used e-mail
 - Browsed the Web
 - Bought a product on the Web (what?)
 - Created a web page using an authoring tool
 - Written some HTML
- ECT 250 will NOT teach you HTML.
- If you have taken CSC 200 see me after class.

Electronic commerce

- To many people the term electronic commerce, often shortened to e-commerce, is equivalent to shopping on the web.
- The term electronic business is sometimes used to capture the broader notion of e-commerce.
- In this course, we will use e-commerce in its broadest sense. It encompasses both web shopping and other business conducted electronically.

E-commerce is not new

- Banks have used electronic funds transfers (EFTs), also called wire transfers, for decades.
- Businesses have been engaging in electronic data interchange for years. EDI occurs when one business transmits computer readable data in a standard format to another business.

Electronic data interchange

- In the 1960s businesses realized that many of the documents they exchanged related to the shipping of goods and contained the same set of information for each transaction.
- By sending the information electronically in a standard format, the businesses could save money on printing, mailing, and re-entry of data.
- Electronic transfer of data also introduces fewer errors than manual transfer.

Technology and commerce

- In order to understand how technology can aid commerce we need to understand traditional commerce.
- Once we have identified what activities are involved in traditional commerce, we can consider how technology can improve them.
- Note that technology does not always improve commerce. Knowing when technology will NOT help is also useful.

Origins of commerce

- The origins of traditional commerce predate recorded history.
- Commerce is based on the specialization of skills. Instead of performing all services and producing all goods independently, people rely on each other for the goods and services they need.
- Example: My mother trades eggs to one of her neighbors in exchange for repairs to the fences on her ranch.

Traditional commerce

- Although money has replaced bartering, the basic mechanics of commerce remain the same: one member of society creates something of value that another member of society desires.
- **Commerce** is a negotiated exchange of valuable objects or services between at least two parties and includes all activities that each of the parties undertakes the complete the transaction.

Views of commerce

Commerce can be viewed from at least two different perspectives:

1. The buyer's viewpoint
2. The seller's viewpoint

Both perspectives will illustrate that commerce involves a number of distinct activities, called ***business processes***.

The buyer's perspective

From the buyer's perspective, commerce involves the following activities:

1. Identify a specific need
2. Search for products or services that will satisfy the specific need
3. Select a vendor
4. Negotiate a purchase transaction including delivery logistics, inspection, testing, and acceptance
5. Make payment
6. Perform/obtain maintenance if necessary

The seller's perspective

From the sellers' perspective, commerce involves the following activities:

1. Conduct market research to identify customer needs
2. Create a product or service to meet those needs
3. Advertise and promote the product or service
4. Negotiate a sales transaction including delivery logistics, inspection, testing, and acceptance
5. Ship goods and invoice the customer
6. Receive and process customer payments
7. Provide after sales support and maintenance

Business processes

Business processes are the activities involved in conducting commerce.

Examples include:

- Transferring funds
- Placing orders
- Sending invoices
- Shipping goods to customers

E-commerce

We will define ***e-commerce*** as the use of electronic data transmission to implement or enhance any business activity.

Example (Figure 1-4): A buyer sends an electronic purchase order to a seller. The seller then sends an electronic invoice back to the buyer.

When used appropriately, electronic transmission can save both time and money.

Impact of e-commerce

E-commerce is changing the way traditional commerce is conducted:

- Technology can help throughout the process including promotion, searching, selecting, negotiating, delivery, and support.
- The value chain is being reconfigured.

Value chain analysis

A way of looking at the activities of an industry or organization.

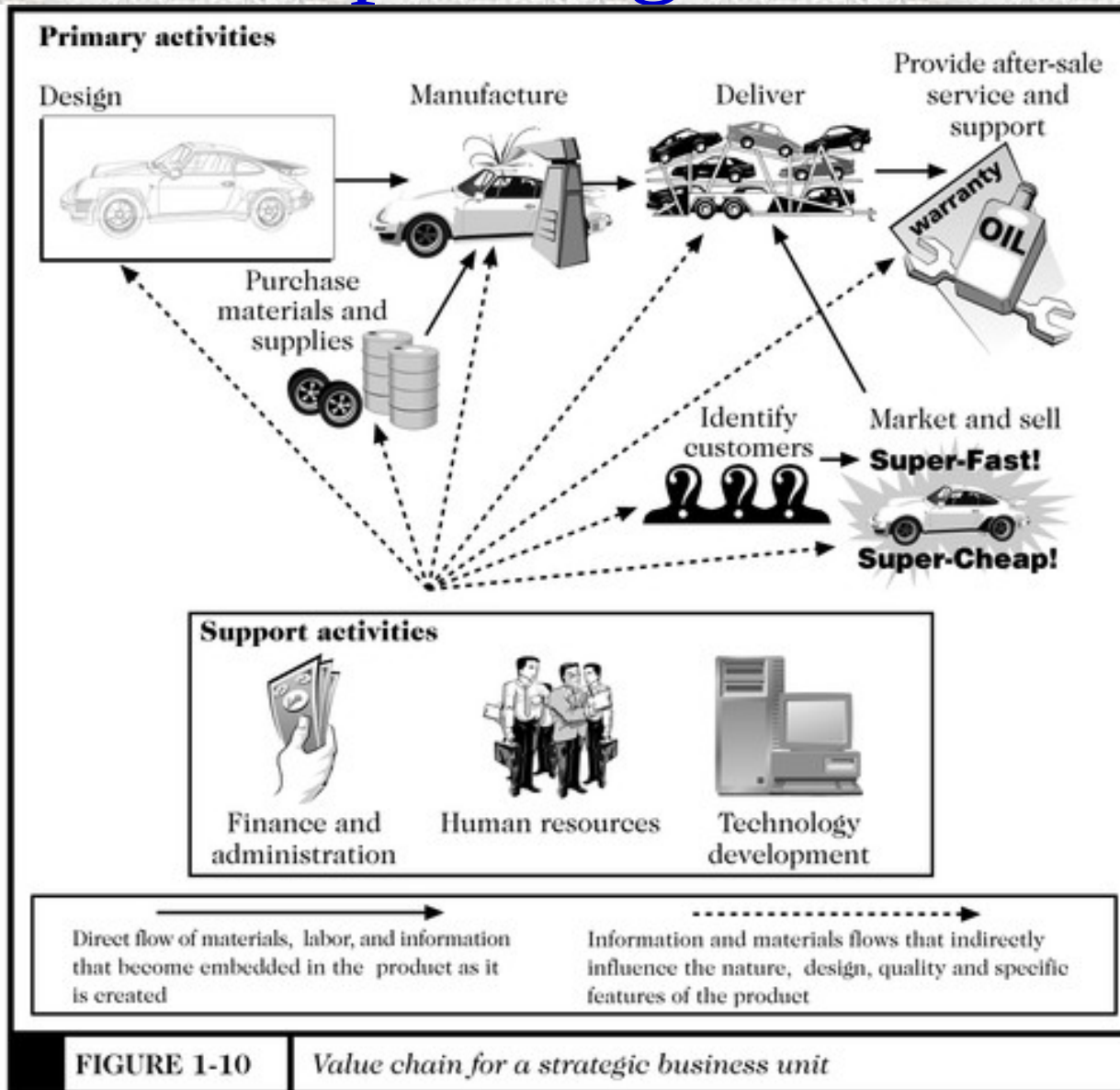
- Primary activities

Costs are directly allocated to a product

- Support activities

Costs are associated with the overall operation of the organization

Example: Figure 1-12



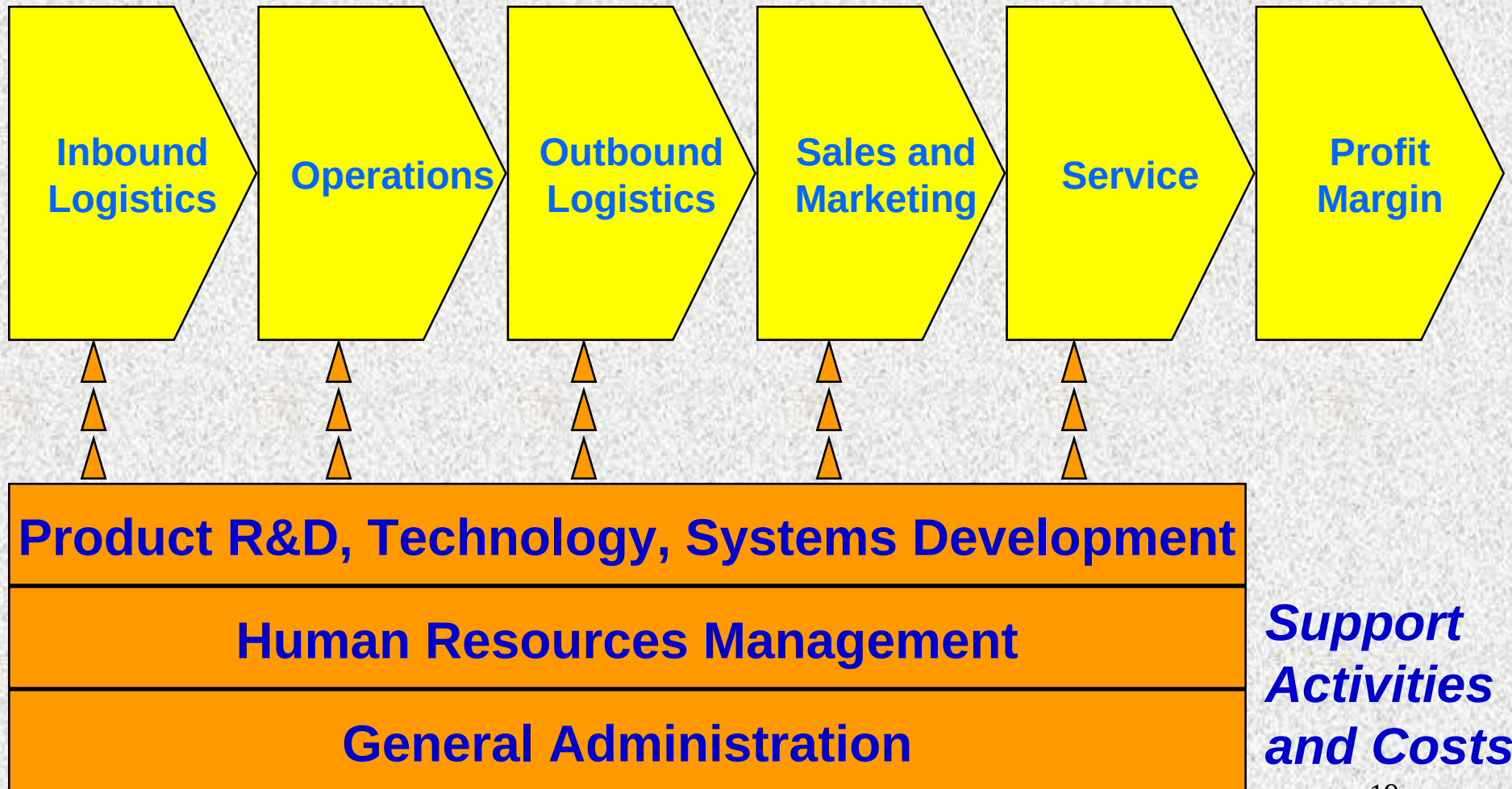
Origin of the name

An organization is a “chain” of activities, each providing a value:

- R&D develops the products
- Sales promotes the products
- Production manufactures the products
- Final inspection assures quality
- Logistics delivers the products
- Technical support maintains the products

A typical company value chain

Primary Activities and Costs



Reconfiguring the value chain

1. *Vertical*

- **Forward integration:** Gaining ownership or increased control over distributors or retailers.

Example: United Airlines buys Priceline

- **Backward integration:** Gaining ownership or increased control over suppliers.

Example: Amazon buys RCA records

2. *Horizontal:* Seeking increased control over competitors.

Example: Amazon buys BestBooksBuys.com

Appropriateness

It is important to identify which business processes can be streamlined using e-commerce technologies.

It is equally important to realize that some processes make effective use of traditional commerce and can't be improved upon using technology.

Technology is not a panacea. Using it when it is not necessary or helpful can be a costly mistake.

Well-suited for e-commerce

Business processes that are well-suited for electronic commerce:

- Sale/purchase of new books and CDs
- Online delivery of software
- Advertising and promotion of travel services
- Online tracking of shipments

The business processes that are especially well-suited to e-commerce include commodity items, that is, a product or service that has become standardized.

Best for traditional commerce

Business processes that are well-suited to traditional commerce:

- Sale/purchase of high fashion clothing (Any possible exceptions?)
- Sale/purchase of perishable food products
- Small-denomination transactions (Future?)
- Sale of expensive jewelry and antiques

In general, products that buyers prefer to touch, smell, or otherwise closely examine are difficult to sell using e-commerce.

Questionable cases

Would e-commerce or traditional commerce work best for the following activities?

- Sale/purchase of rare books
- Browsing through new books
- Sale/purchase of shoes
- Sale/purchase of collectibles (trading cards, plates, etc.)

Combinations of both

Some business processes can be handled well using a combination of electronic and traditional methods:

- Sale/purchase of automobiles
- Online banking
- Roommate-matching services
- Sale/purchase of investment/insurance products

In this course we will discuss the issue of evaluating the advantages and disadvantages of e-commerce.

Let's consider a few examples now.

Advantages of e-commerce

For the seller:

- Increases sales/decreases cost
- Makes promotion easier for smaller firms
- Can be used to reach narrow market segments

For the buyer:

- Makes it easier to obtain competitive bids
- Provides a wider range of choices
- Provides an easy way to customize the level of detail in the information obtained

Advantages of e-commerce II

In general:

- Increases the speed and accuracy with which businesses can exchange information
- Electronic payments (tax refunds, paychecks, etc.) cost less to issue and are more secure
- Can make products and services available in remote areas
- Enables people to work from home, providing scheduling flexibility

Disadvantages of e-commerce

- Some business processes are not suited to e-commerce, even with improvements in technology
- Many products and services require a critical mass of potential buyers (e.g. online grocers)
- Costs and returns on e-commerce can be difficult to quantify and estimate
- Cultural impediments: People are reluctant to change in order to integrate new technology
- The legal environment is uncertain: Courts and legislators are trying to catch up